

Team Final Report

Testing 1 to 60

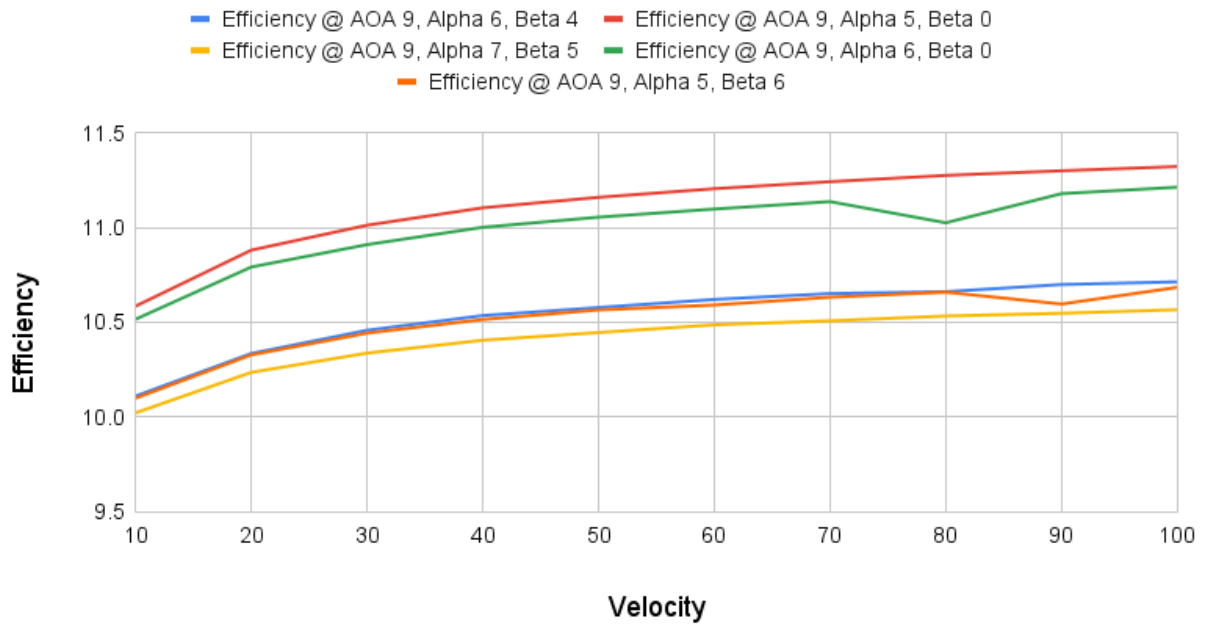
Team-Name: Subsonic

Sr. No.	PDC Number	Prototype Name (CAD Name)
1	4542	PCR03-THW-MOF-PXE6
2	4543	PCR03-THW-MOF-PXE6
3	4544	PCR03-THW-MOF-PXE6
4	4545	PCR03-THW-MOF-PXE6
5	4547	PCR03-THW-MOF-PXE6

Official Remark:

Team-Name: Subsonic

Efficiency vs Velocity



1. By observing the team excel sheet graph, which curve is at the highest position? For which geometrical condition are you getting the highest value of efficiency?

Observation Report Based on Efficiency vs Velocity Graph

Upon reviewing the provided Efficiency vs Velocity graph, the curve that consistently occupies the highest position across the entire velocity range is the red-colored curve, which corresponds to the geometrical condition:

AOA = 9° , Alpha = 5° , Beta = 0° .

Key Observation

- This configuration demonstrates the highest efficiency values throughout the tested velocity range (10 m/s to 100 m/s).
- The efficiency under this condition reaches approximately 11.1–11.3, which is higher than all other plotted configurations.
- The trend indicates a steady increase in aerodynamic performance with increasing velocity, without any significant dips compared to other curves.

Conclusion

The geometrical configuration of Angle of Attack (AOA) = 9° , Alpha = 5° , and Beta = 0° provides the maximum aerodynamic efficiency among all tested settings. This implies that minimizing beta deflection while maintaining moderate alpha leads to optimal airflow management and reduced drag losses, thereby enhancing the overall efficiency of the wing/flap system.

2. By observing the team excel sheet, if we move from best performing to worst performing curve, what is the effect of different AOA conditions on that trend?

Efficiency Ranking of Conditions (Best → Worst)

Based on overall efficiency levels across velocities:

a. Alpha 5, Beta 0 (PDC4542) – Highest efficiency (~10.58 → 11.32)

- This configuration consistently delivers the best aerodynamic efficiency.
- Efficiency increases steadily with velocity.
- Lower Beta appears to provide smoother airflow and lower drag.

b. Alpha 6, Beta 0 (PDC4544) – High efficiency (~10.51 → 11.21)

- Also strong performance, slightly below Alpha 5 Beta 0.
- Increasing Alpha from 5 → 6 slightly increases lift but increases drag enough to reduce efficiency marginally.

c. Alpha 6, Beta 4 (PDC4545) – Moderate efficiency (~10.11 → 10.71)

- Introduction of Beta = 4 leads to lower efficiency compared to Beta = 0 cases.
- Indicates increased sideslip or asymmetric flow effects increasing drag.

d. Alpha 5, Beta 6 (PDC4543) – Moderate efficiency but less stable (~10.09 → 10.68)

- Similar to Alpha 6 Beta 4 but shows more fluctuations at high velocities (drop at 90 m/s).
- Higher Beta affects flow stability.
- e. Alpha 7, Beta 5 (PDC4547) – Lowest efficiency (~10.02 → 10.56)

- Highest Alpha and high Beta produce highest drag, reducing aerodynamic efficiency.
- Efficiency improves with velocity, but remains lower than all other sets.

Trend Observed from Best to Worst Performing Curves

As we move from the best-performing curve (Alpha 5 Beta 0) to the worst-performing curve (Alpha 7 Beta 5):

- Efficiency decreases overall.
- Higher Alpha increases lift but increases drag disproportionately, lowering efficiency.
- Higher Beta worsens performance due to increased lateral forces and induced drag.
- Combinations involving higher Beta values (4, 5, 6) consistently reduce aerodynamic smoothness.

Velocity Influence

- Efficiency increases with velocity, but
- The rate of improvement is highest for low-Beta conditions and smallest for high-Beta conditions.

Key Observations

- Beta = 0 cases always outperform Beta 4–6 cases.
- Lower Alpha (5°) provides the optimum balance between lift and drag.
- High Alpha + high Beta (Alpha 7, Beta 5) significantly reduces aerodynamic efficiency.
- Efficiency differences between curves become more apparent at high velocities.

Conclusion

The data clearly shows that the best aerodynamic performance occurs at lower Alpha and zero Beta, while higher Alpha and higher Beta lead to reduced efficiency due to increased drag and destabilized airflow. The transition from best to worst curves demonstrates how sensitive aerodynamic efficiency is to AOA variations, especially Beta-induced flow asymmetry.

3. By observing the team excel sheet, if we move from best performing to worst performing curve, what is the effect of different Beta conditions on that trend?

If we rank the curves based on overall efficiency (highest → lowest), the order is approximately:

1. $\beta = 0^\circ, \alpha = 5^\circ$ – *Best performing*
2. $\beta = 0^\circ, \alpha = 6^\circ$ – *Second best*
3. $\beta = 4^\circ, \alpha = 6^\circ$ – *Middle performance*
4. $\beta = 6^\circ, \alpha = 5^\circ$ – *Lower performance*
5. $\beta = 5^\circ, \alpha = 7^\circ$ – *Worst performing*

Effect of Beta on Performance Trend

1. Lower Beta ($\beta \approx 0^\circ$) gives highest efficiency

- The two best-performing curves are $\beta = 0^\circ$ (Red and Green).
- This indicates that reducing the outer flap deflection (Beta) improves aerodynamic efficiency.
- Lower Beta causes less drag and maintains a more optimal pressure distribution.

2. Moderate Beta ($\beta = 4^\circ$ – 5°) decreases efficiency

- As Beta increases from $0^\circ \rightarrow 4^\circ \rightarrow 5^\circ$, efficiency drops consistently.
- The Blue ($\beta=4^\circ$) and Yellow ($\beta=5^\circ$) curves show:
 - Slightly reduced efficiency at all velocities.
 - Flattened efficiency growth with velocity.
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3. Higher Beta ($\beta \approx 6^\circ$) gives the lowest efficiency

- The worst-performing curve (Orange) corresponds to $\beta = 6^\circ$.
- Larger flap deflection increases:
 - Induced drag
 - Flow separation risk
 - Losses in lift-to-drag ratio
- This visibly reduces the curve's efficiency compared to all others.

Overall observation

As the Beta flap angle increases, the efficiency of the wing consistently decreases because higher Beta introduces more drag and disrupts the airflow over the rear section of the wing. Therefore, the curves clearly show that lower Beta settings (especially $\beta = 0^\circ$) give the best aerodynamic efficiency, while higher Beta angles result in the poorest performance.

4. By observing the team excel sheet, if we move from best performing to worst performing curve, what is the effect of different Alpha conditions on that trend?

Analysis of Performance Trend

The curves were ranked based on their **Average Efficiency** across the velocity range. The rankings are as follows:

Rank	Performance	Alpha	Beta	Average Efficiency
1	Best	5	0	11.11
2	High	6	0	10.99
3	Medium	6	4	10.54
4	Low	5	6	10.51
5	Worst	7	5	10.41

Effect of Alpha Conditions

Observing the trend from the **best-performing** to the **worst-performing** curve reveals a distinct relationship with the Alpha conditions:

1. **Best Performance (Alpha 5):** The highest efficiency is achieved at the lowest Alpha setting (**Alpha 5**), specifically when combined with a low Beta (Beta 0).
2. **Intermediate Performance (Alpha 6):** As we move down the performance ranking, the Alpha value increases to **Alpha 6**. For example, even with the same Beta of 0, increasing Alpha from 5 to 6 causes a drop in efficiency (from 11.11 to 10.99).

3. **Worst Performance (Alpha 7):** The worst-performing curve corresponds to the highest Alpha setting (**Alpha 7**). Even when compared to a curve with a "worse" (higher) Beta (Alpha 5, Beta 6), the Alpha 7 curve performs more poorly, highlighting the strong negative impact of high Alpha on efficiency.

Conclusion

Moving from the best-performing to the worst-performing curve corresponds to an increase in the Alpha value.

The data indicate a clear inverse relationship between Alpha and performance efficiency. The **best performing curve** is defined by the lowest Alpha condition (**Alpha 5**), while the **worst performing curve** is defined by the highest Alpha condition (**Alpha 7**). Although the Beta variable also influences the results (where a higher Beta reduces efficiency), the trend dictates that maintaining a lower Alpha is critical for maximising performance. Therefore, to optimise for the best trend, one should prioritise minimising the Alpha angle.